

App Dev Project Report

1. Student Details

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About me: I am interested in Web Development and Artificial Intelligence . Alongside my tech pursuits, I am actively working on laying the foundation for an innovative agriculture business. I thrive at the intersection of modern technology and sustainable real-world enterprise.

2. Project Details (for MAD1)

Project Title: Trekking Management Application

Problem Statement:

Adventure organizations require efficient systems to manage trekking activities involving trek organizers, staff, and participants. Currently, many trekking groups rely on spreadsheets, phone calls, or manual coordination, which makes it difficult to manage trek approvals, track bookings, avoid overbooking, and maintain trek history

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows organizers to manage trek schedules efficiently and provides users with a dashboard to book treks and track history.

3. AI/LLM Declaration

I used Gemini to assist in drafting documentation, optimizing SQLAlchemy query syntax, and debugging error logs during development. The extent of AI/LLM usage is around 5–10%, limited to syntax suggestions. All final database models, routing structure, and business logic were implemented manually.

Category	Ai used	notes
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Backend (Flask + DB)	7 - 10 %	Mostly for syntax error
Frontend (HTML)	2-4 %	Mostly for syntax error in css
Styling/UI	0%	
Debugging	2-3%	For understanding new unknown word in error

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
Flask-SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend UI styling and responsive design
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data
flask-Bcrypt	For hashing password
datetime	For save data as a date

5. Database Schema / ER Diagram

Tables:

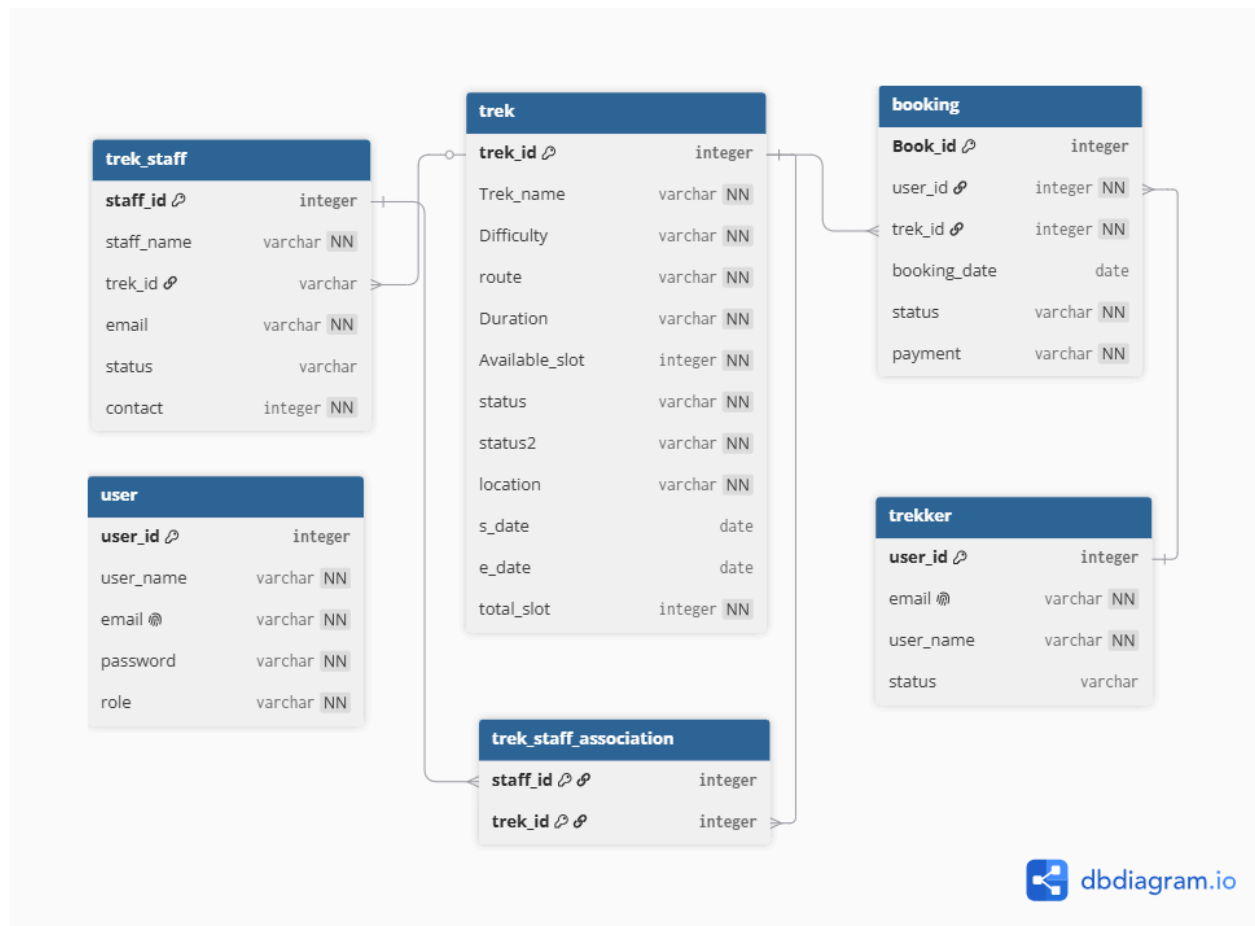
1. User- store all trekker and trek staff email password role
2. Trek- store all details about trek
3. Trek_staff - store all information about trek staff only
4. Trekker- store all information about user(trekker) only

5. Booking-have all booking details done by users
6. Trek_staff_association-it is association table

Relationships:

one-to-Many: trek→booking, trekker→booking

Many-to-Many: trek→trek staff



6. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **/controller.py** - it has all flask route and logic
- **/model.py**- database Models using flask sqlalchemy
- **/database.py** - database initiate
- **Templates file**- have all html file
- **Static file** - have css file

Implemented Features:

All core features like login ,signin, dashboard management, booking, history etc .

7. Video Presentation

DriveLink:

https://drive.google.com/file/d/1XfxuxCiiF3VxDfXpe1i08N_eN28-g4Oh/view?usp=sharing

Note: however video presentation was over 10min long that why i cut rest of last few min, using inshort.